

Don't Pop It! Modifying the Balloon Analogue Risk Task for a Course-Based Research Experience

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Abstract

Background: Relatively few studies focus on Course-Based Undergraduate Research Experiences (CUREs) to improve retention of knowledge and enhance student research experience in the social sciences.

Objective: We introduced a modified, manual version of the Balloon Analogue Risk Task (BART), in which students engaged in all phases of a research project, including the development of a risk-taking assessment, participant recruitment, scale administration, presentation of the modified BART, data collection, and data entry and analysis.

Method: To determine the effectiveness of the CUREs, students self-reported their research knowledge and skills using pre-post measures. Additionally, we measured attitudes and perceptions of the student's experience with the research project.

Results: Results from the learning outcomes demonstrated increased research knowledge and skills and a positive overall experience with the project.

Conclusion: The modified BART is an effective CURE for engaging undergraduate psychology students enrolled in research methods courses.

Teaching Implications: Engagement through hands-on research project design and data collection, as demonstrated with the modified BART, may serve to increase the overall retention of information and enhance class experience, potentially creating a pipeline for further engagement in high-impact practices for undergraduate psychology students.

Keywords

Course-Based Undergraduate Research Experience, research methods, statistics, high-impact practices, Balloon Analogue Risk Task

Course-Based Undergraduate Research Experiences (CUREs) represent an approach to instruction that incorporates research experiences into the classroom. Research on CUREs suggests many benefits to students who engage in these classroom research projects (see [Corwin et al., 2015](#), for a list of benefits in Life Science courses). These include greater motivation in science and greater interest in graduate studies ([Brownell et al., 2012](#)). Perhaps most importantly, it allows students from underrepresented groups to gain research experiences that they may otherwise have not known about or had access to ([Banger & Brownell, 2014](#)). For these reasons and others, [Kuh \(2008\)](#) has identified engagement in research as a high-impact practice for undergraduate students.

Despite their popularity in the physical sciences, CUREs and research on their effectiveness in the social sciences are lacking ([Ruth et al., 2023](#)). We are particularly interested in assessing the effectiveness of CUREs given various barriers to implementing individualized research experiences (e.g., limited opportunities to engage with research-active faculty). CUREs allow students to participate in a more intensive research experience akin to a traditional undergraduate research experience (URE) in the typical class setting while reducing additional burdens on teachers outside the classroom. The current study is an example of an easy-to-administer, small-scale CURE that was embedded in a psychology research methods course.

Specifically, students engaged in a project to examine whether a modified, manual version of the Balloon Analogue Risk Task (BART; [Lejuez et al., 2002, 2003](#)) would predict risk-taking behaviors in college students similarly to the computerized version. At the beginning and end of the semester, students enrolled in the course were surveyed about their research knowledge and skills; at the end of the semester, they reported on their perceptions of the project, its usefulness, and their future interest in research.

The Balloon Analogue Risk Task

Risk-taking behavior is assessed via various experimental and nonexperimental methods, including measures related to impulsive behaviors and faulty decision-making, as well as self-reports on riskiness. A study by [Lejuez et al. \(2002\)](#) validated

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these approaches by assessing the relationship between self-reported risky behavior and performance on the BART. The BART is a computerized measure of risk-taking behavior in which participants are instructed to inflate a virtual balloon as much as possible via individual key presses that simulate a single balloon pump. To incentivize the task, each pump carries with it a 5-cent reward, which can be collected at any time, yet the money is lost if the balloon pops. In the original BART study, participants also completed a series of self-report risk assessments alongside various other measures (IQ, personality, depression, anxiety, and addiction). Of relevance to the current study, measures of risk-related constructs positively correlated with BART performance (i.e., greater number of balloon pumps), suggesting that self-reports of risk-taking behavior and this experimental risk measure converged on the same construct (Lejuez et al., 2002).

Student Involvement in the BART

As detailed below, to further assess the construct validity of the BART, students in the class administered a manual version of the BART to other students on campus alongside a brief self-report measure of risk-taking behaviors. We chose to modify the original computerized BART to a manual version so that students could be more engaged with data collection.

Students engaged in all project design and data collection phases, including survey development, participant recruitment, scale administration, and presentation of the modified BART. First, students completed their human subject certification in the research ethics module, allowing them to collect and analyze data. Next, in groups of five to seven, students generated constructs of interest that could be related to risk-taking. After narrowing down the list of constructs as a class, students, once again in groups, generated several questionnaire items for each construct, using principles of survey development covered in class. The groups were not necessarily comprised of the same students for each group exercise. Many of these questions were added to risk-taking scale items used in the original BART study (Lejuez et al., 2002).

For data collection, students were self-selected into research roles, with some opting to recruit participants while others administered the survey or manual BART. Several students were assigned to measure the balloons and helped coordinate all research activities.

After data collection, numerous students entered data in preparation for analysis. The data collected by the students were subsequently analyzed in a series of in-class workshops covering descriptive statistics and correlation and regression analyses. For instance, to assess the validity of the manual BART and brief risk-taking assessment, students correlated the diameter of the balloon with the composite assessment score, with the prediction that larger balloon sizes would be positively associated with higher self-report risk scores.

The modified BART project presented an ideal conceptual replication design for the research methods course. It engaged students in a quantitative correlational research project that

included the development of a brief risk-taking assessment and a hands-on observational measure (i.e., balloon inflation). Beyond correlation and regression analyses, this research experience highlighted important psychometric constructs, including validity (e.g., convergent validity: correlation between the self-reported risk scores and the BART metric or balloon size) and reliability (e.g., inter-rater reliability: correlation between student raters' measures of balloon size).

Student Learning Outcomes

To assess student learning outcomes, we introduced a pre-post measure of research experience, adapted from Kardash (2000), to determine if the project contributed to self-reported gains in research knowledge and skills. Student's understanding of concepts related to psychological research methods and statistical analyses was assessed at the beginning of the semester. At the end of the semester, students completed this measure again, along with a survey about their attitudes toward and perceptions of the research experience. We predicted an increase in self-reported research knowledge and skills for the pre-post measure, along with positive ratings of the overall research experience for the post-BART attitudes and perceptions measure.

Method

Participants

Participants were students enrolled in a research methods ($N = 48$) course at a large state university in Metropolitan Atlanta. Although specific demographic information was unavailable for the course, students were all Psychology majors, typically in their second or third year. The University is a majority-minority institution, with 51% of students receiving Pell grants. This is the first course in a required two-course sequence in the Psychology major. The course covers fundamental topics in psychological research design, such as ethics and forms of validity, specifically focusing on descriptive research designs (observational and survey research). Students are also exposed to multiple statistical analyses, including chi-square, one-sample t -test, significance testing with bivariate correlations, and simple regression. The entire class syllabus is available with the Open Science Framework (OSF) materials (Kelly & Parrish, 2024). At the end of the semester, $n = 37$ students (77%) consented to have their data used for further analysis. The BART research project itself and data collection on the CURE were approved in separate applications by the IRB at Georgia State University (Protocols #H24097 & H24071).

Materials and Procedure

The Balloon Analogue Risk Task. Data collection occurred on two separate afternoons outside of class time. Students enrolled in the course recruited participants in the lobby of a large campus building. On average, the research team consisted of

about 13 students on any given day. Students selected one of the data collection roles: Recruiter/Survey administrator (5), Balloon Administrator (5), Balloon Measurer (2), and Data Coordinator (1).

The recruiters approached potential participants in the lobby, asking if they would be willing to fill out a survey and inflate a balloon for a course project. They were also enticed by the possibility of a Starbucks gift card to inflate the largest balloon. After agreeing to participate and signing the informed consent document, participants answered a 29-item measure of risk-taking. In addition to some demographic questions, participants responded to items on risk-taking from pre-existing scales, including the *Barratt Impulsiveness Scale* (Steinberg et al., 2013), *Brief Sensation Seeking Scale* (Hoyle et al., 2002), and the *UPPS Impulsive Behavior Scale* (Whiteside & Lynam, 2001), as well as items generated by students about constructs of interest (e.g., tattoos, competition, etc.). Sample items included the following: “I like to do frightening things,” “I am not a very cautious person,” and “I have trouble controlling my impulses.”

Following the administration of the risk-taking scale, participants were introduced to the Balloon Administrators. These individuals gave participants a balloon and a hand pump and instructed participants to manually inflate the balloon to the largest desired size without popping it. Participants were reminded that the two participants with the largest balloons would receive gift cards. When participants indicated they were done inflating their balloons, students helped tie the balloons. Next, two balloon measurers recorded (in centimeters) the diameter of the balloon with a soft tape measure around the widest part of the balloon. The average diameter of the two ratings was used as the participant’s BART score (inter-rater reliability: $r = .98$). If the balloon popped during inflation, the experimenter recorded it, and the participant became ineligible for the prizes. In total, 99 participants completed the risk-taking survey and balloon inflation task. Materials related to the modified BART (script, measures, data sheets, lab assignments), as well as a student presentation of the correlational results, are available in the OSF materials (Kelly & Parrish, 2024).

Student Learning Outcome Measures. Research methods students completed two scales assessing learning outcomes. One measure was a 19-item measure of self-reported research knowledge and skills (adapted from Kardash, 2000), completed online during the first whole week of class and then again in class during the second to last week of class. In this survey, participants rated their agreement to statements on a scale of 1 (*strongly disagree*) to 5 (*strongly agree*) (see Table 1).

The second measure was a 12-item post-BART measure of overall experience completing the research project, using the same 1 through 5 scale (see Table 2). This was completed in class and during the second-to-last week of class (but during a separate class period than the post-test content knowledge measure).

Results

Self-Reported Research Knowledge and Skills

We analyzed the pre-post questionnaire to determine if the research experience was associated with self-reported improvement of knowledge and skills related to the course. Specifically, we conducted paired-samples t -tests using a Bonferroni-corrected alpha level ($\alpha = .003$). Effect sizes were calculated using Cohen’s d and interpreted according to traditional conventions (Cohen, 1992). The results of the analysis can be seen in Table 1. Items are presented in descending order based on effect size. Average scores on all but one item significantly increased from the beginning of the semester to the end of the semester (all t s > 2.11 , p s $< .02$). The items with the largest differences as indexed by effect size concerned being able to statistically analyze data, interpreting correlations, and relating data to the original hypothesis. Importantly, these activities were conducted with data and examples stemming from the BART project. The item with the smallest effect size pertained to communication ability. Given that this project did not explicitly involve a presentation of the results in an oral or written form, this result also makes sense.

BART Experience Attitudes and Perceptions Survey

At the end of the semester, students completed a survey on their experience with the research project. We conducted a one-sample t -test, Bonferroni corrected for multiple comparisons ($\alpha = .004$) to determine if the students’ perceptions of the project were significantly more positive than the neutral (i.e., 3) option on the 1 to 5 scale. The results can be seen in Table 2. Students agreed that participation in the BART project helped them learn important research design concepts (e.g., operational definitions and writing survey items) and increased knowledge and confidence in conducting statistical calculations (all t s > 5.93 , all p s $< .001$). Students strongly agreed that participating in the BART project increased their interest in conducting and participating in more research. There also was strong agreement that the project should be retained in future iterations of the class, suggesting that students found the project enjoyable and worth completing again.

Discussion

The current study evaluates a CURE example in a research methods course. Undergraduate psychology students in the course worked together to plan, develop, and ultimately conduct a research study regarding whether self-reported attitudes and behaviors around risk-taking could be predicted by the size a person is willing to inflate a balloon (an adapted manual BART task, modified from Lejuez et al., 2002). Students then analyzed the data using bivariate correlations and simple regression. The results from a pre- and post-test measure of self-reported research knowledge and skills suggested that this project may have aided in students

Table 1. Performance on Pre- and Post-Measures of Self-Reported Research Knowledge and Skills.

Item	Pre M(SD)	Post M(SD)	<i>t</i>	<i>p</i>	<i>d</i>
Statistically analyze data	2.13 (1.07)	3.97 (0.89)	8.53	< .001*	1.56
Interpret correlation results (e.g., the correlation coefficient and <i>p</i> -value)	2.13 (1.04)	3.83 (0.99)	7.22	< .001*	1.32
Interpret data by relating results to the original hypothesis	2.66 (1.04)	4.03 (0.73)	6.15	< .001*	1.14
Understand concepts related to psychological research methods (e.g., ethical standards, research design, data analysis)	2.47 (1.1)	4.01 (0.69)	6.13	< .001*	1.12
Contribute to data collection for a psychology experiment	2.83 (1.18)	4.47 (0.78)	5.64	< .001*	1.03
Identify a specific question for investigation based on the research in psychology	2.93 (1.08)	4.13 (0.68)	5.41	< .001*	0.99
Confidently complete a statistics assignment	2.70 (1.30)	3.92 (0.73)	5.08	< .001*	0.98
Adequately prepare for a statistics exam	2.37 (1.28)	3.70 (0.72)	4.81	< .001*	0.93
Write survey items for a psychology experiment	2.7 (1.29)	3.8 (0.81)	4.65	< .001*	0.85
Do well on a final exam in a statistics course	2.67 (1.27)	3.63 (0.63)	4.20	< .001*	0.81
Formulate a research hypothesis based on a specific question	3.37 (1.07)	4.23 (0.57)	4.07	< .001*	0.74
Interpret the meaning of a Results table in a journal article	2.52 (1.09)	3.56 (0.89)	3.64	< .001*	0.70
Design an experiment to test a research hypothesis	2.87 (1.11)	3.67 (0.71)	3.69	< .001*	0.67
Understand the importance of “controls” in research	3.6 (1.13)	4.43 (0.73)	3.54	< .001*	0.65
Think critically about psychological research	3.22 (1.28)	4.07 (0.62)	3.35	.001*	0.65
Relate results to the “bigger picture” in psychology	3.22 (1.31)	4.11 (0.80)	3.13	.002*	0.60
Understand the statistics in a journal article	2.52 (1.37)	3.48 (0.85)	2.87	.004	0.55
Make use of the primary scientific research literature in psychology (e.g., journal articles)	2.67 (1.24)	3.47 (0.82)	2.98	.003	0.54
Orally communicate the results of research projects	3.26 (1.35)	3.89 (0.75)	2.11	.022	0.41

Note. The scale minimum is 1 (*strongly disagree*), and the maximum is 5 (*strongly agree*). Standard deviations are presented in parentheses. The effect size was measured with Cohen's *d*. * indicates statistically significant differences.

understanding of the material. These data are also consistent with answers from the BART Experience Survey, in which students reported general agreement that the project helped them learn many of these concepts. This survey also suggested that students enjoyed the research experience, which motivated them to engage in future research projects.

In our view, this project offers several benefits. One advantage of this specific project is that the modified BART was intuitive, interesting, and relatively easy for students to administer, generating a large sample size ($N=99$) across only two data collection sessions. Furthermore, the materials needed to run the experiment were all low-cost (e.g., balloons, balloon pumps, tape measures, and gift cards in small amounts). Another major advantage of this project is that it allows instructors to include various topics commonly covered in psychological research methods courses. The ability to use a single study as a recurring example in the class was a real positive. Returning to the same project ideas throughout the semester can create a sense of continuity, which may help deepen student learning. Additionally, using the modified BART project offered opportunities to reflect on previously covered topics that have not been actively discussed. For example, correlational analyses failed to show a significant relationship between vaping and self-reported risk-taking. This led to an interesting discussion about principles of survey wording to examine how we might have asked that question differently (e.g., asking about smoking instead of vaping). Similarly, given that balloon size did not regularly correlate with measures

of risk-taking, the class discussed issues of experimental control, replication, and confounding or extraneous variables that may have influenced people's behaviors. For instance, did the presence of other people (i.e., *observer bias*) cause participants to inflate their balloons more than they would have if they were alone?

Accordingly, this CURE can serve as a pipeline to more traditional UREs conducted within a faculty member's lab, particularly enhancing access to such opportunities among historically underrepresented groups (NASEM, 2017). Beyond exposing students to a career path, they may not have known existed, this initial experience may demystify the research process. The sense of competency students may derive from the experience gives them more confidence to pursue further research experiences. A large-scale review of UREs among STEM disciplines outlined several benefits to students engaging in UREs, including but not limited to enhanced technical skills, better understanding of the research process, enhanced readiness for future careers, increased graduation rates, and an increase in intrinsic motivation in the subject matter (Buchanan & Fisher, 2022).

At many universities, class sizes are quite large, and there are more majors than opportunities for research experiences, especially UREs. Smaller, teaching-centered colleges may offer fewer opportunities for individually mentored student research because of high teaching loads. A CURE, as described here, offers every student enrolled in the course an opportunity to experience all portions of the research experience, which can

Table 2. Students Self-Reported Ratings Towards the BART Experience.

Item	Mean rating	t-Value	d
Understanding various statistical topics (e.g., random sampling, data collection, and statistical analysis)	4.18 (0.72)	10.22*	1.64
Creating operational definitions	3.89 (0.82)	6.83*	1.09
Understanding the importance of writing survey items	4.08 (0.81)	8.33*	1.33
Understanding of the ethical treatment of human subjects	4.26 (0.79)	9.99*	1.60
Knowledge of statistical analyses used in psychological science	3.97 (0.84)	7.22*	1.16
Confidence in conducting statistical analyses	3.87 (0.83)	6.54*	1.05
Understanding elements of statistical analyses (e.g., means, degrees of freedom, p-values)	3.87 (0.86)	6.30*	1.01
Investment in learning how to conduct statistical analyses	4.00 (0.88)	7.03*	1.13
Interest in conducting and participating in more psychological research at the University	4.28 (0.86)	9.34*	1.50
Understanding how experimental psychology could be used to assess other topics in psychology	4.14 (0.75)	9.19*	1.50
I would recommend retaining the hands-on class project in future offerings of the class	4.78 (0.67)	16.14*	2.65

Note. The scale minimum is 1 (*strongly disagree*), and the maximum is 5 (*strongly agree*), with a midpoint value of 3 (*neutral*). Standard deviations are presented in parentheses. The effect size was measured with Cohen's *d*. (*) = $p < .001$.

serve as a basis for determining whether research activity is something they wish to pursue. CUREs like the modified BART can be helpful because they provide a meaningful research experience for students while not placing additional burdens on teachers outside the classroom. Like UREs, the literature on CUREs suggests they can increase content knowledge and analytic skills, yield higher student retention, motivation, and self-efficacy, and enhance student confidence in their science skills (see Buchanan & Fisher, 2022 for a comprehensive review).

A final appealing aspect of this CURE is that it could easily be modified to meet the needs of the course. This project would still be useful in programs where statistics and research methods are separate courses. In a pure statistics course, data from this project could be used to conduct many different analyses. In the current project, we focused on correlations and simple regression, but the data could be used to demonstrate other tests. For instance, chi-square tests can be used to examine the association between balloon pops and gender or balloon size and balloon color. One could collect demographic data on academic standing and analyze whether self-reported riskiness differs across years in school with an analysis of variance.

In a pure research methods course, instructors could focus more on the study's planning. We had students work on developing operational definitions of constructs by writing survey questions and taking a correlational approach; however, this project could be modified to be part of an experimental manipulation. Students could investigate particular manipulations that might make risk-taking more likely (e.g., balloon color) and use balloon inflation as the primary risk-taking indicator. In programs where students take these courses in sequence, often in the same calendar year, this project could help serve as a mechanism through which students can identify the important connections between research design and statistical analysis. Finally, instructors could use a project like this to demonstrate open science principles by pre-registering their project at the beginning of the semester and sharing data. If time permits, students could also be asked to complete writing and presentation assignments related to this project. Students could reflect on

their experiences in the research study and be asked to write up a short empirical paper or design a poster and present the results in a conference-like setting (see Frankowski, 2023 for other suggestions). Indeed, a small group of students from the course and students who participated in a pilot version of the study presented this work as a poster at institutional and regional research conferences, thus further expanding the reach of the project.

A key limitation of the study is the lack of a control group or a course section, which prevented this research experience from being included. This would allow us to unequivocally report that the increases in learning and student engagement observed in the pre-post design were due to the inclusion of this high-impact practice above and beyond the regular course content. Future course iterations may include a comparison group; however, the pre-post design provides promising support for the content gained from implementing BART, including evidence of student engagement in research both within and outside the classroom. In short, the manual version of the BART test appears to be a fun and interesting CURE that can help solidify class knowledge and skills, at least as far as student self-perceptions are concerned, and it increases interest in research among undergraduate psychology students. Furthermore, it is an easy-to-implement, intuitive, and low-cost project design that can be adapted for various topics and class sizes.

Data Availability

All raw data for the student learning outcomes are freely available online at https://osf.io/7naj9/?view_only=f61413cc3253449d972c7aa23ed41efa.

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